

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

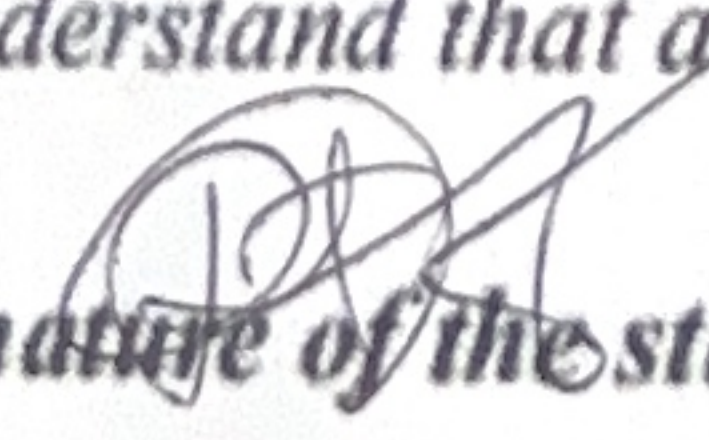
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Ravi Teja Y.V (192565533) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

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AP

Problem 1: Logistics Company - Backtracking Search

Constraints

B_1 Can't be assigned To Route C.

only one Bus per route.

B_3 Can't be assigned To Route A.

Every bus must be assigned exactly one route

Step 1: Assign $B_1 \rightarrow$ Route A

Step 2: $B_2 \rightarrow$ Route B

$B_1 \rightarrow$ Route A

$B_2 \rightarrow$ Route B

Step 3: $B_3 \rightarrow$ Route C

check Constraints:

- B_1 is not in Route C
- B_2 is Before B_3 .
- one Bus per route.
- B_3 is not Route A
- Every bus assigned once

Final Valid Schedule

Bus	Route
B ₁	Route A
B ₂	Route B
B ₃	Route C

Backtracking:
The Algorithm assign buses one by one. After each assignment it checks all constraints. Every assignment satisfies, no backtracking is required. If failed. The algorithm would undo the last assignment and try another route.

Problem 2: Campus delivery Robot:

- a) • The Robot can move only in four direction up, down, left, right.
- The robot can observe only adjacent cell, so it has incomplete kn to the environment.
- Normal movement cost 1, while pedestrian zones cost 3 (1+2 extra).

- Since the environment is partially observable, it updates its map while moving.

b) Solution using uniform cost search

Initial State: Start at S

Goal at G

Step 1: Place start node in Priority Queue with cost = 0.

Step 2: Expand the node with smallest cost.

Step 3: Ignore blocked construction cell.

Step 4: update the robot knowledge after observing neighbouring cells.

Step 5: Continue expanding the lowest-cost node.

eg: $S \rightarrow A = 1$

$A \rightarrow B = 1$

$B \rightarrow C = 3$

$C \rightarrow D = 1$

$D \rightarrow G = 1$

Total = 7

c) AI Concept Used:

- The delivery robot acts as an intelligent agent.

- It Processes the best action.

- It moves toward the goal.

Environment type:

- Partially observable (Can't see Wholemap).
- Dynamic (New info is discovered while moving).
- Discrete (moves between grid cells).
- Sequential (each action affects future

decision).

D) Justification:

- It always find the minimum - cost path
- It avoids blocked construction zones.
- It handles different movement costs effectively.
- It Works well in partially observable environment by updating knowledge during navigations.
- It helps the intelligent agent make rational decision and safely reach the goal.

COURSE NAME: Artificial Intelligence

OUTCOME COVERED

Apply AI basics such as intelligent agents to solve problems effectively. (BL3)

Tool Weightage: 50%

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Annexure

Analytical P

(Name / Registered to the guidelines violation of academic

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